

SCHOOL OF COMPUTER ENGINEERING

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

PROJECT PROPOSAL

QualityGuard AI

Deep Learning-Based Industrial Product Defect Detection and Quality Inspection System

Submitted By

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Abstract

Industrial quality inspection is an essential part of modern manufacturing, where identifying defective products accurately and at an early stage can reduce production losses, improve product quality, and minimize dependence on manual inspection. Traditional visual inspection methods are often time-consuming, subjective, and difficult to scale for high-volume manufacturing environments.

This project proposes **QualityGuard AI**, a deep learning-based computer vision system designed to automatically identify defects in industrial product images. The proposed system will use image preprocessing and transfer learning techniques to classify products into predefined quality categories such as normal and defective, with the possibility of further classification into specific defect types such as scratches, cracks, dents, or surface irregularities.

Pre-trained convolutional neural network architectures such as MobileNetV2, EfficientNet, or ResNet will be investigated and fine-tuned using an industrial defect image dataset. The models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix. Grad-CAM-based visual explanations will also be explored to highlight image regions that contribute to the model's prediction.

A lightweight web-based interface will be developed using Gradio, allowing users to upload an industrial product image and receive the predicted defect category, confidence score, and quality assessment. The system will also provide an automatically generated inspection report.

Keywords: Computer Vision, Deep Learning, CNN, Transfer Learning, Industrial Defect Detection, Image Classification, MobileNetV2, EfficientNet, Grad-CAM.

1. Introduction

Manufacturing industries rely heavily on quality inspection to ensure that products meet required standards before reaching customers. In conventional manufacturing environments, visual inspection is frequently performed manually by trained workers. Although manual inspection can be effective, it can become inefficient when production volume is high, inspection must be performed continuously, or defects are small and difficult to identify consistently.

Recent developments in Artificial Intelligence and Computer Vision have enabled machines to analyze images and identify visual patterns automatically. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated strong performance in image classification and visual inspection tasks.

The proposed project, **QualityGuard AI**, aims to develop an automated image-based quality inspection system capable of analyzing industrial product images and identifying whether a product is normal or defective. The system will use transfer learning from a pre-trained CNN architecture, reducing computational requirements and training time compared with training a deep neural network completely from scratch.

The project will also investigate explainability. Instead of producing only a classification result, Grad-CAM will be used to visualize the regions of an image that influenced the model's prediction.

2. Problem Statement

Manual visual inspection in industrial manufacturing has several limitations:

- It can be time-consuming for large-scale production.
- Inspection results may vary between individual inspectors.
- Small surface defects can be difficult to identify consistently.
- Continuous inspection requires significant human resources.
- Manual inspection becomes difficult to scale with increasing production volume.
- Maintaining inspection records manually can be inefficient.

Therefore, there is a need for an automated computer vision system capable of analyzing product images and identifying potential defects efficiently and consistently.

QualityGuard AI addresses this problem by developing a deep learning-based image classification system for automated industrial visual inspection.

3. Aim

The primary aim of this project is to develop an AI-powered industrial visual inspection system that uses deep learning and computer vision to detect and classify defects in product images.

The proposed system will provide:

1. Automated product image analysis.
2. Defect classification.
3. Prediction confidence.
4. Model performance evaluation.
5. Explainable predictions using visualization techniques.
6. A user-friendly web interface.
7. Automated quality inspection reports.

4. Objectives

The major objectives of the project are:

1. To collect and prepare a suitable industrial defect image dataset.
2. To perform image preprocessing including resizing, normalization, and data augmentation.
3. To develop a baseline image classification model.
4. To investigate transfer learning using pre-trained CNN architectures such as MobileNetV2, EfficientNet, and ResNet.
5. To compare the performance of different deep learning models.
6. To identify a suitable model considering both classification performance and computational requirements.
7. To evaluate the final model using accuracy, precision, recall, F1-score, and confusion matrix.
8. To investigate Grad-CAM or a similar explainability method for visualizing important regions of an image.
9. To develop a web-based interface for image upload and prediction.
10. To generate a structured inspection report containing the input image, prediction, confidence score, and quality status.

5. Proposed System

QualityGuard AI will consist of the following major components:

1. Image Input Module
2. Image Preprocessing Module
3. Deep Learning Classification Module
4. Prediction and Confidence Module
5. Explainability Module
6. Web Interface
7. Inspection Report Generation Module

The proposed system will follow the workflow shown below.

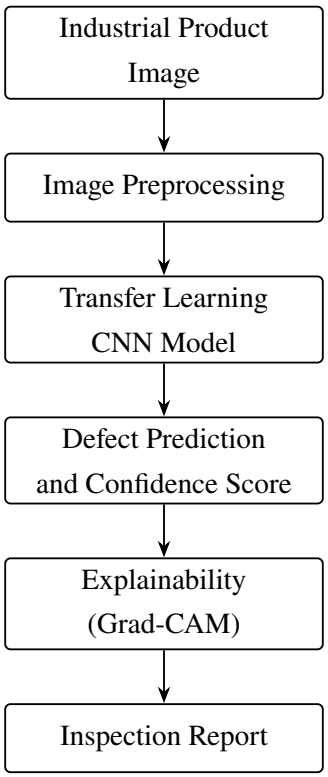


Figure 1: Proposed system workflow

6. Methodology

The project will follow a structured deep learning workflow consisting of dataset preparation, preprocessing, model development, training, evaluation, explainability, and application development.

6.1 Dataset Collection

A publicly available industrial defect image dataset will be selected for the project. The dataset will contain images representing normal and defective products or surfaces.

Depending on the selected dataset, defect categories may include:

- Normal / defect-free surface
- Scratch
- Crack
- Dent
- Surface irregularity

- Corrosion or similar defects

The final classes will depend on the selected dataset and its suitability for the project.

6.2 Image Preprocessing

The image preprocessing pipeline will include:

- Image resizing according to the selected CNN architecture.
- Pixel normalization.
- Handling of corrupted or unsuitable images.
- Training, validation, and testing split.
- Data augmentation such as rotation, flipping, zooming, and translation where appropriate.

Data augmentation will be used to improve model generalization and reduce overfitting.

6.3 Model Development

The project will investigate transfer learning using pre-trained CNN architectures.

Candidate architectures include:

- MobileNetV2
- EfficientNet
- ResNet

A pre-trained model will initially be used as a feature extractor, followed by a task-specific classification layer. Selected layers may subsequently be fine-tuned using the industrial defect dataset.

The general architecture will be:

Input Image → Pre-trained CNN → Feature Extraction → Global Pooling → Dense Layer → Classification

6.4 Model Training

The dataset will be divided into training, validation, and testing subsets.

The models will be trained using an optimizer such as Adam. Categorical cross-entropy or binary cross-entropy will be used depending on the classification formulation.

Early stopping and learning-rate scheduling may be used to improve training efficiency and reduce overfitting.

6.5 Model Evaluation

The trained models will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC, where applicable

For multi-class classification, macro and weighted averages will also be considered.

6.6 Explainable AI

Grad-CAM will be investigated to improve the interpretability of the model.

Grad-CAM produces a heatmap highlighting regions of an input image that contribute significantly to the model's prediction.

This will help demonstrate where the model identified visual evidence associated with a particular defect.

6.7 User Interface

A lightweight web interface will be developed using Gradio.

The interface will allow a user to:

1. Upload an industrial product image.
2. Run the trained model.
3. View the predicted defect class.
4. View the prediction confidence.
5. View the Grad-CAM visualization.
6. Generate an inspection report.

7. System Architecture

The architecture of QualityGuard AI is divided into five major stages: input acquisition, image preprocessing, model inference, explainability, and report generation.

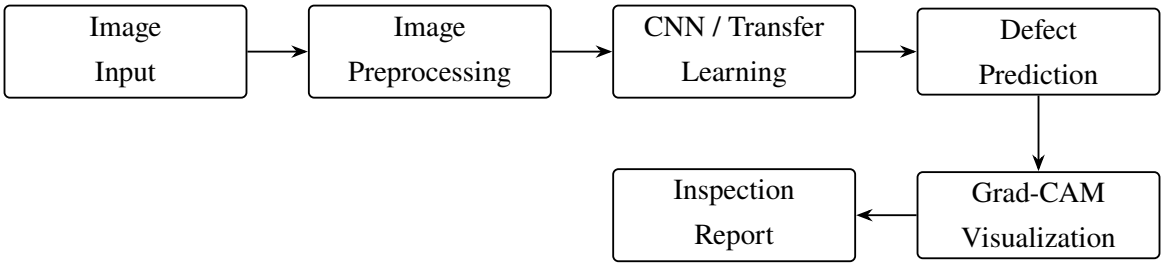


Figure 2: System architecture of QualityGuard AI

8. Tools and Technologies

Technology	Purpose
Python	Primary programming language
TensorFlow / Keras	Deep learning model development
OpenCV	Image preprocessing and computer vision operations
NumPy	Numerical computation
Pandas	Dataset handling and analysis
Matplotlib	Data visualization and evaluation plots
Scikit-learn	Data processing and evaluation metrics
MobileNetV2 / EfficientNet / ResNet	Transfer learning architectures
Grad-CAM	Model explainability
Gradio	Web-based user interface
FPDF / ReportLab	Inspection report generation
Jupyter Notebook / VS Code	Development environment
Git / GitHub	Version control and collaboration

Table 1: Proposed technologies and their applications

9. Dataset Requirements

The project requires an image dataset containing industrial products or surfaces with clearly labelled quality conditions or defect categories.

The selected dataset should ideally contain:

- Multiple defect categories.
- Sufficient images per category.
- Consistent image labels.
- Images suitable for transfer learning.
- Sufficient variation in image appearance.

The final dataset and number of classes will be finalized during the initial implementation phase based on data quality, class balance, and project feasibility.

10. Expected Outcomes

At the completion of the project, the following outcomes are expected:

1. A trained deep learning model capable of classifying industrial product images.
2. A comparison of multiple transfer-learning architectures.
3. Quantitative evaluation of the selected model.
4. Confusion matrix and other model performance visualizations.
5. Explainable predictions using Grad-CAM.
6. A functional Gradio-based web interface.
7. Automated generation of an industrial inspection report.
8. A complete demonstration of an AI-assisted visual quality inspection workflow.

11. Proposed Contribution

The main contribution of QualityGuard AI is the integration of deep learning, computer vision, explainable AI, and application development into a single industrial inspection workflow.

The proposed system combines:

- Deep learning-based defect classification.

- Confidence-based prediction.
- Explainable AI visualization.
- Image upload and prediction interface.
- Automated inspection reporting.

This combination aims to provide a practical prototype for AI-assisted visual quality inspection.

12. Scope of the Project

The initial scope of the project includes image-based classification of industrial product defects using a publicly available dataset.

The first version will focus on offline image analysis rather than direct integration with industrial machinery or real-time production-line cameras.

Future extensions may include:

- Real-time camera-based inspection.
- Object detection for locating multiple defects.
- Image segmentation for precise defect boundaries.
- Edge-device deployment.
- Integration with manufacturing execution systems.
- Automated production-line monitoring.
- Continuous model improvement using newly collected inspection data.

13. Expected Limitations

The performance of the proposed system may depend on:

- Dataset size and quality.
- Class imbalance.
- Lighting and background variations.
- Similarity between different defect categories.
- Image resolution.

- Available computational resources.

The proposed system will therefore be considered an AI-assisted inspection tool rather than a complete replacement for established industrial quality-control procedures.

14. Project Timeline

The initial development and prototype implementation of QualityGuard AI is planned over a period of one month.

Week	Planned Activities
Week 1	Dataset selection and collection, exploratory data analysis, image preprocessing, dataset organization, and project environment setup.
Week 2	Implementation of baseline CNN and transfer-learning models, training of MobileNetV2/EfficientNet/ResNet models, and initial hyperparameter tuning.
Week 3	Model evaluation and comparison using accuracy, precision, recall, F1-score, confusion matrix, and implementation of Grad-CAM-based explainability.
Week 4	Development of the Gradio interface, integration of the final model, automated inspection report generation, system testing, documentation, and preparation of the final presentation.

Table 2: One-month project development timeline

15. Feasibility Study

The proposed QualityGuard AI system is technically and practically feasible within the planned project duration.

15.1 Technical Feasibility

The project will use established deep learning frameworks and publicly available datasets. Transfer learning will be used to reduce training time and computational requirements. The system can be developed using Python, TensorFlow/Keras, OpenCV, and Gradio.

The project does not require specialized industrial hardware during the initial development stage. A standard computer with a suitable GPU or a cloud-based computational environment will be sufficient for model training and experimentation.

15.2 Operational Feasibility

The proposed system is designed with a simple image-upload interface so that users can submit product images and obtain an automated inspection result. The interface will require minimal technical knowledge and will present the prediction, confidence score, and visual explanation in an understandable format.

15.3 Economic Feasibility

The project will primarily use open-source software, frameworks, and publicly available datasets. Therefore, the development cost is expected to remain low. No specialized hardware or commercial machine-learning software is required for the initial prototype.

15.4 Time Feasibility

The initial prototype is planned to be completed within one month. The work will be divided into dataset preparation, model development, evaluation, application development, and final testing.

16. Team Responsibilities

The project will be developed collaboratively by all six members. Responsibilities will be distributed according to the major components of the system.

Team Member	Primary Responsibility
Priyanshu Aryan Panda	Project coordination, ML pipeline, model development, system integration, and final implementation.
Swayamjit Sahoo	Dataset collection, preprocessing, exploratory data analysis, and dataset organization.
Sakshi Sahoo	Deep learning model experimentation, transfer learning, training, and tuning.
Biswajeet Behera	Model evaluation, performance analysis, confusion matrix, and visualization.
Lohitakhya Sahu	Gradio interface development, user interaction, and application integration.
Srutiman Mohapatra	Grad-CAM explainability, report generation, testing, documentation, and presentation support.

The responsibilities are not strictly isolated. All members will participate in system integration, testing, documentation, and final presentation activities.

17. Project Deliverables

The following deliverables are planned for the completion of the project:

1. **Industrial Defect Dataset:** A cleaned and organized dataset suitable for training and evaluating the proposed deep learning models.
2. **Preprocessing Pipeline:** A reusable image preprocessing and augmentation pipeline.
3. **Trained Deep Learning Models:** One or more transfer-learning-based CNN models trained for industrial defect classification.
4. **Model Evaluation:** Performance analysis using accuracy, precision, recall, F1-score, and confusion matrix.
5. **Explainable AI Module:** Grad-CAM-based visualization showing the image regions contributing to the prediction.
6. **QualityGuard AI Application:** A functional Gradio-based application allowing users to upload product images and obtain predictions.
7. **Inspection Report:** An automatically generated report containing the input image, predicted defect, confidence score, and inspection result.
8. **Project Documentation:** Complete project documentation containing the methodology, experiments, results, limitations, and future scope.
9. **Final Presentation:** A presentation demonstrating the problem, methodology, model performance, application, and results.

18. Conclusion

QualityGuard AI proposes a practical application of deep learning and computer vision for automated industrial quality inspection. The system will analyze product images and identify potential defects using transfer-learning-based CNN architectures.

The project combines machine learning, computer vision, explainable AI, and application development into a unified industrial inspection workflow. By providing prediction confidence, visual explanations, and automated inspection reports, the proposed system aims to demonstrate how AI can assist quality-control processes in manufacturing environments.

The proposed one-month development plan focuses on building a functional and experimentally validated prototype while keeping the implementation technically feasible. The system can subsequently be extended toward real-time inspection, object detection, segmentation, and integration with industrial production systems.